



电子科技大学

University of Electronic Science and Technology of China

求 / 实 / 求 / 真

大 / 气 / 大 / 为

2025 MCM ---F题经验分享

团队成员：王翊钊，赵浩安，周逸轩
汇报人：王翊钊，赵浩安



目录/Contents

1. 题目分析
2. 建模求解
3. 备赛建议
4. 赛中建议

01 题目分析

如何从题目中提取赛题重点，如何建立赛题思路



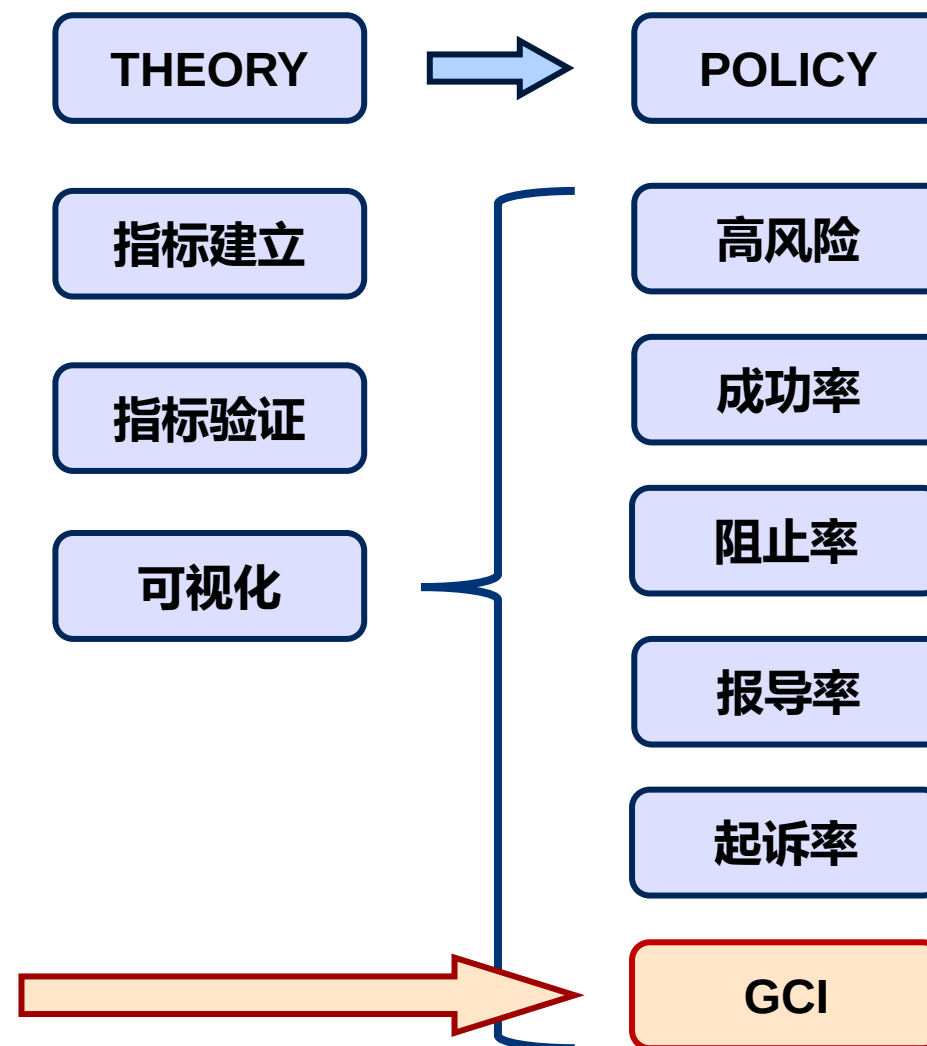
赛题分析

Requirements:

In this problem, you are asked to help identify patterns that could inform the data-driven development and refinement of national cybersecurity policies and laws based on those that have demonstrated effectiveness. **Develop a theory for what makes a strong national cybersecurity policy and present a data-driven analysis to support your theory.** In developing and validating your theory, things you may wish to consider include:

- How is cybercrime **distributed** across the globe? Which countries are disproportionately **high targets of cybercrimes**, where are **cybercrimes successful**, where are **cybercrimes thwarted**, where are **cybercrimes reported**, where are cybercrimes **prosecuted**? Do you notice any patterns?
- ~~As you explore the published **national security policies** of various countries and compare these with the distribution of cybercrimes, what **patterns emerge** that would help you identify parts of a policy or law that are particularly effective (or particularly ineffective) in addressing cybercrime (through prevention, prosecution, or other mitigation efforts)? Depending on your analytical approach, it may be relevant to consider **when each policy was adopted**.~~
- What **national demographics** (e.g., access to internet, wealth, education levels, etc.) **correlate** with your cybercrime distribution analysis? And how might these support (or conflate with) your theory?

Your work should not seek to create a new measure of cybersecurity, as there are existing measures such as ITU's Global Cybersecurity Index (GCI),^[1] which assigns a score to each country based on their level of cybersecurity as assessed through five pillars: legal, technical, organizational, capacity building, and cooperation. Instead, you have been asked to seek meaningful patterns in the effectiveness of national cybersecurity policies and/or laws with respect to the national contexts in which those policies were enacted. **The GCI or similar existing research may be useful in validating your work.** Additional resources that could be useful include websites that collect cybercrime data, particularly those leveraging the VERIS framework, which attempts to standardize how cybercrime data is collected and reported,^[2] including the VERIS Community Database (VCDB).^[3] You are encouraged to find other data sources but be mindful of the veracity and completeness of those sources.

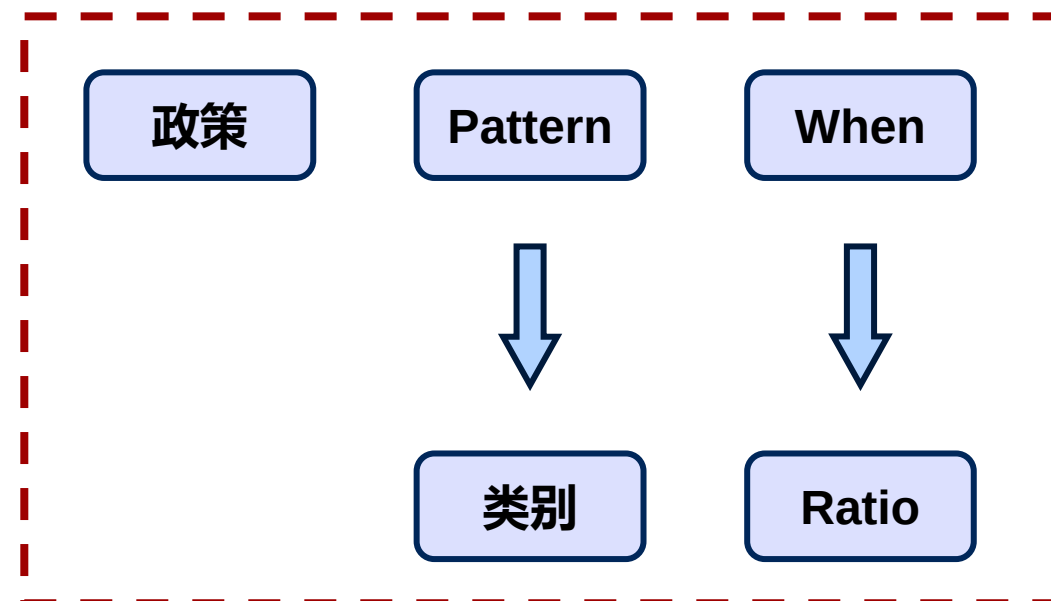
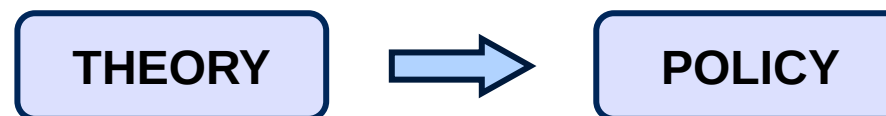


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Share Your Insights:

Use your work to create a 1-page memo to country leaders (nontechnical policy experts) attending an upcoming ITU Summit on Cybersecurity. This memo should provide a nontechnical overview of your work, including a summary of the objective and context, your theory, and the most pressing findings that would be relevant to this audience of national policy-makers.

THEORY



POLICY

人口统计特征

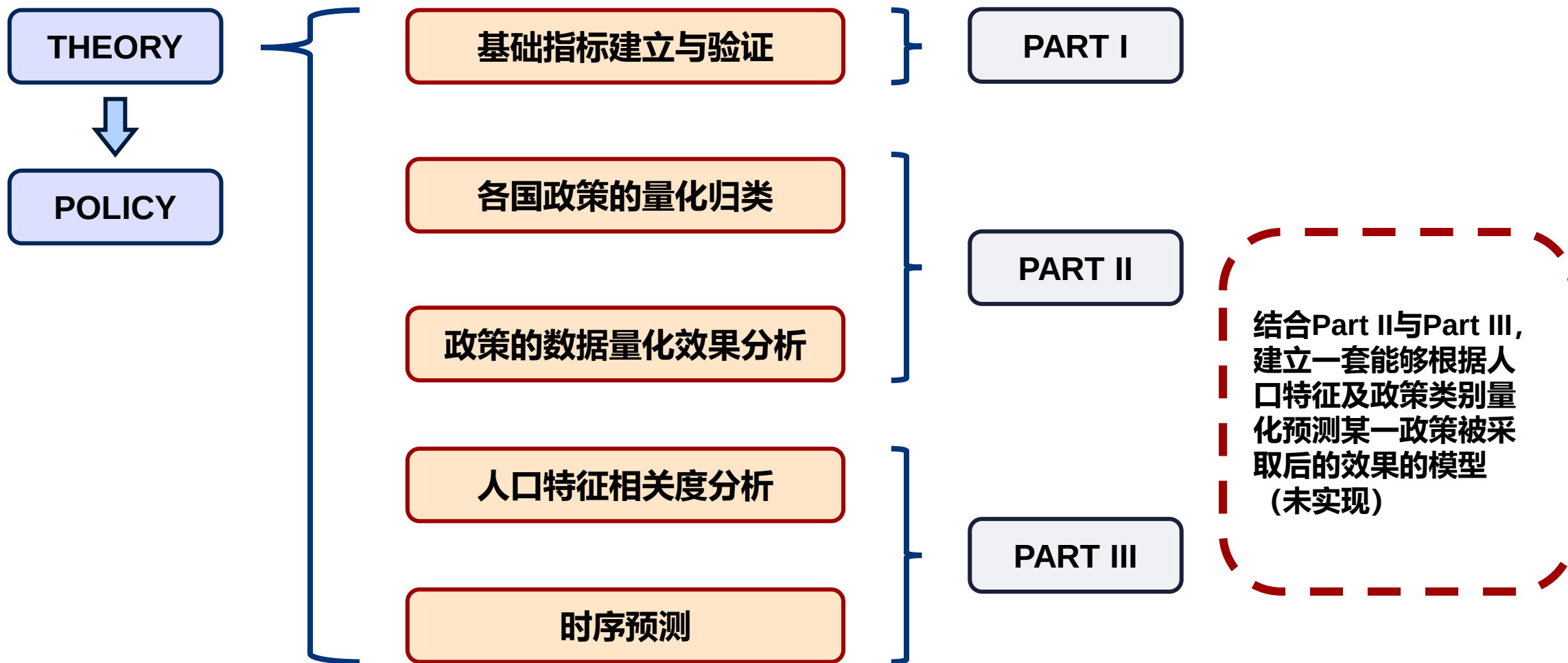
相关性分析

时序预测

Memo

非技术性

思路总结



02

建模求解

根据思路总结的3个Part对模型进行求解



数据预处理

数据可视化

指标建立

指标验证

数据集来源: <https://github.com/vz-risk/VCDB/tree/master>

重点关注: Readme文件 Index

- vcdb_diff.json - An update to the verisc.json schema
- vcdb_diff-labels.json - An update to the verisc-labels.json
- vcdb.json - The vcdb schema file
- vcdb-labels.json - The vcdb labels file
- vcdb-merged.json - The full schema, combining vcdb.json and vcdb-labels.json
- vcdb-enum.json - A json file containing just the enumerations from the schema.
- vcdb-keynames-real.txt - A text file containing the keys in the vcdb schema.

```
{
  "$schema": "http://json-schema.org/draft-04/schema#",
  "additionalProperties": false,
  "description": "VERIS VCDB Schema 1.4.0.",
  "properties": {
    "incident_id": {
      "type": "string",
      "description": "<a href='http://veriscommunity.net/incident-track.html' target='_blank'>More Info</a>"
    },
    "security_incident": {
      "type": "string",
      "description": "Confirmed incident?"
    },
    "confidence": {
      "type": "string",
      "enum": [
        "High",
        "Medium",
        "Low",
        "None"
      ]
    }
  }
}
```

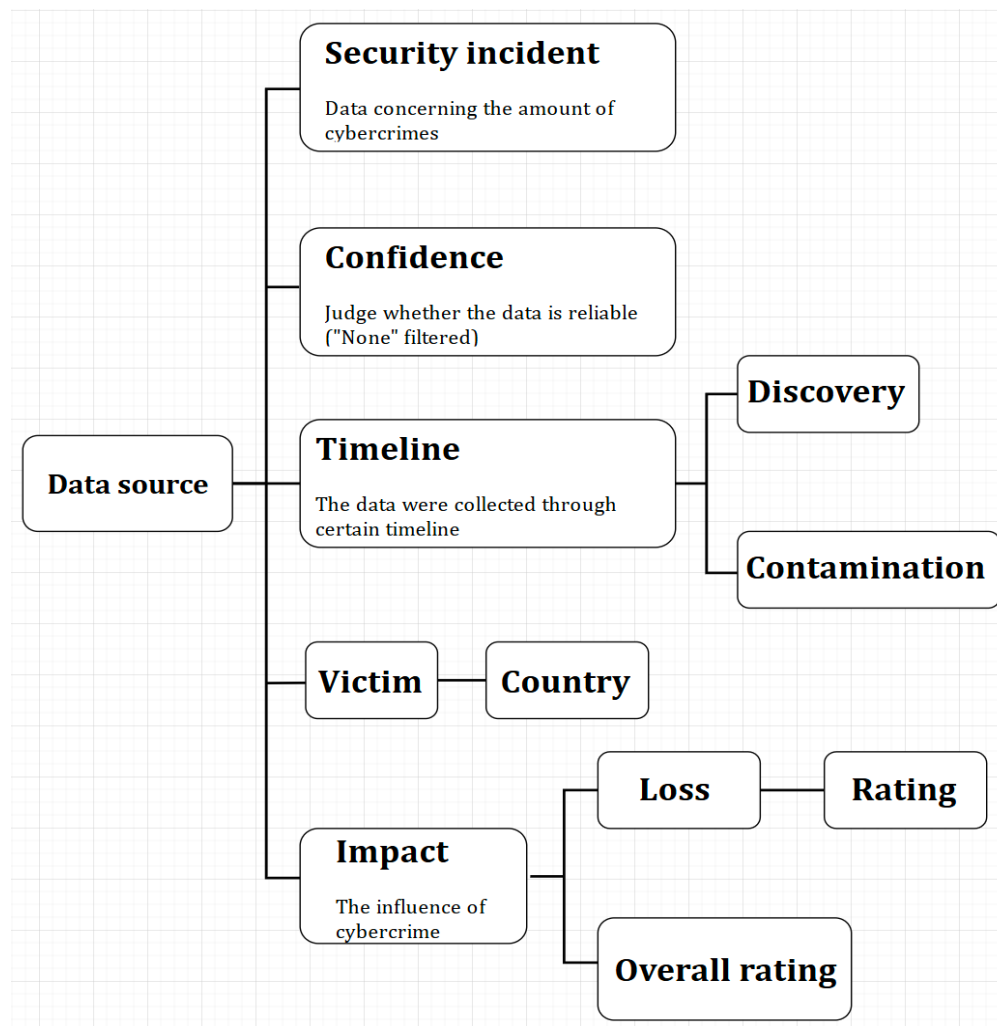
Part I

数据预处理

数据可视化

指标建立

指标验证



Part I

数据预处理

数据可视化

指标建立

指标验证

清洗前文件示例
(部分)
总计122行

```
{
  "action": {
    "misuse": {
      "variety": [
        "Privilege abuse"
      ],
      "vector": [
        "LAN access"
      ]
    }
  },
  "actor": {
    "external": {
      "country": [
        "Unknown"
      ],
      "motive": [
        "Financial"
      ],
      "region": [
        "000000"
      ],
      "variety": [
        "Unknown"
      ]
    },
    "internal": {
      "motive": [
        "Unknown",
        "Financial"
      ],
      "variety": [
        "Other"
      ]
    }
  },
  "asset": {
    "assets": [
      {
        "variety": "S - Database"
      }
    ],
    "cloud": [
      "Unknown"
    ]
  }
}
```

清洗后文件示例
(全部)
总计19行

```
{
  "timeline": {
    "incident": {
      "month": 5,
      "year": 2012
    }
  },
  "victim": {
    "country": [
      "US"
    ],
    "state": "FL"
  },
  "impact": {
    "overall_rating": "Unknown"
  },
  "security_incident": "Confirmed",
  "confidence": "NA"
}
```

Part I

数据预处理

数据可视化

指标建立

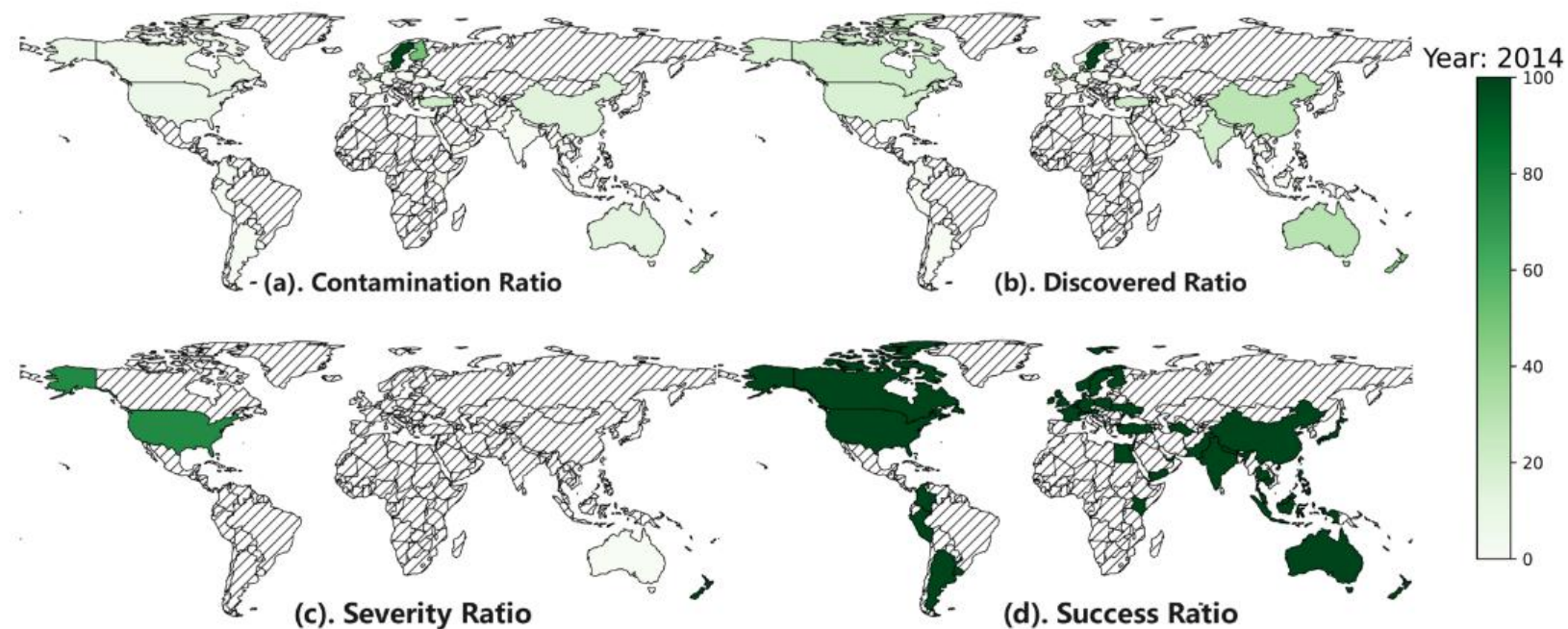
指标验证

$$\text{Ratio of being found} = n_{dc|it}/n_{it} \quad (1)$$

$$\text{Ratio of being thwarted } r_{th|it} = n_{con|it}/n_{it} \quad (2)$$

$$\text{Ratio of causing loss } r_{loss|it} = \frac{\sum_{n=0}^{n_{it}} X_{loss|nit}}{n_{it}-n_{s|it}} \quad (3)$$

$$\text{Severity} = \frac{\sum_{n=0}^{n_{it}} X_{sig|nit}}{n_{it}-n_{or|it}} \quad (4)$$



Part I

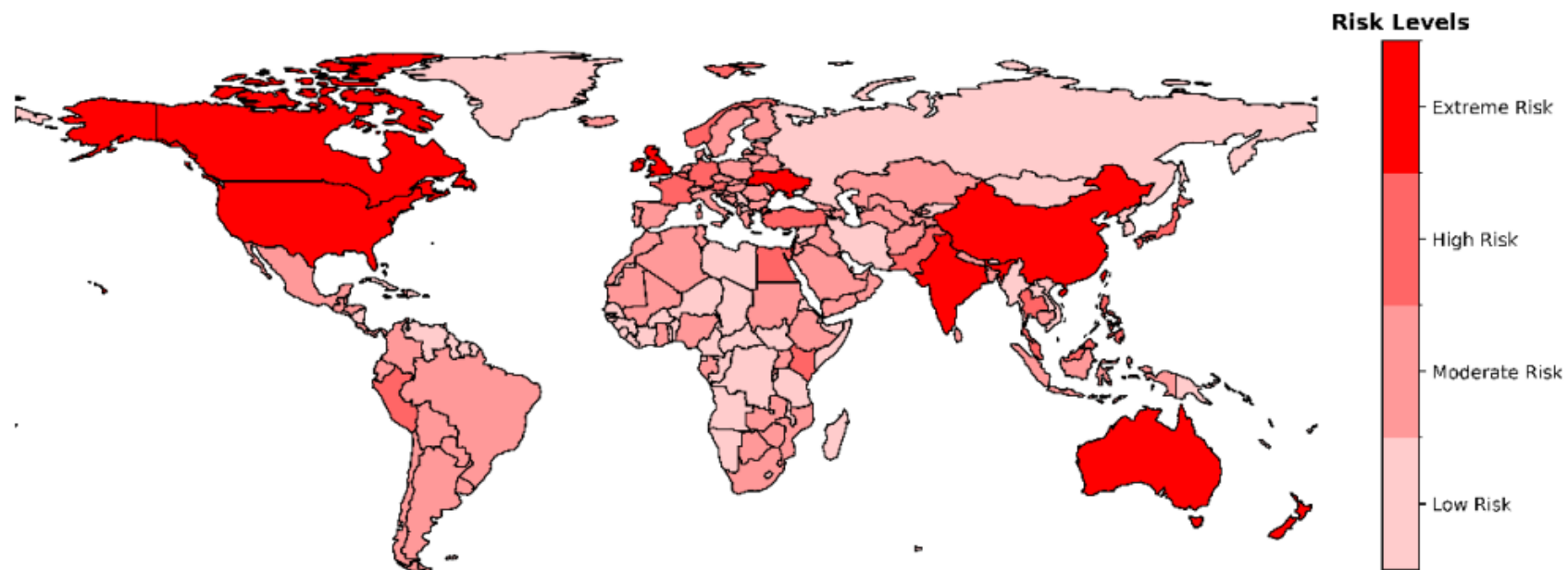
数据预处理

数据可视化

指标建立

指标验证

$$\overline{n_{eff|it}} = \frac{\sum_{t_{start}}^{t_{end}} n_{it}(1-n_{con|it})}{t_{end}-t_{start}} \quad (5)$$



(a) Global risk distribution in 2014

Part I

数据预处理

数据可视化

指标建立

指标验证

GCI来源: <https://www.itu.int/en/ITU-D/Cybersecurity/Pages/global-cybersecurity-index.aspx>

Forth iteration

2020 Global Cybersecurity Index (v4). [GCI Fourth Edition Report](#) (العربية, 中文, English, Français, Русский, Español)

Third iteration

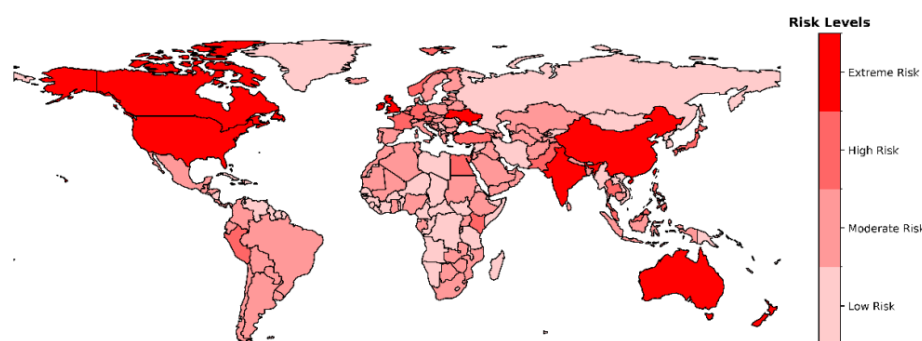
2018 Global Cybersecurity Index (v3). [Download Report GCI v3 \(2018\)](#).

Second iteration

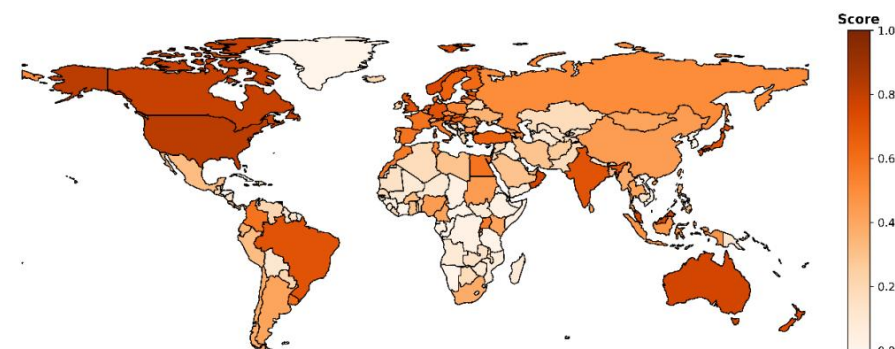
2017 Global Cybersecurity Index (v2). [Download Report GCI v2 \(2017\)](#).

First iteration

2014 Global Cybersecurity Index (v1). [Download Report GCI v1 \(2014\)](#).



(a) Global risk distribution in 2014



(a) GCI distribution in 2014

Part II

政策收集

怎么选?

哪些国家的数据多?更方便分析出内在规律?

根据数据拐点锁定年份, 重点查该年前后政策。

对国家进行分层:地区、收入水平, 排除主观性。

政策分析

数据处理

怎么读?

先关注政策的来源:发布机构?是战略还是法律?

重点关注是发布日期还是生效日期, 确保严谨。

找到文件中的特色内容:创立专门机构, 建设基础设施, 避免泛泛而谈:培养民众意识。

Part II

政策收集

政策分类

制度、多部门合作、公众教育与人才培育。

政策强度区分：战略/处罚，需要不同权重。

政策作用区分：预防，执法，恢复。用来解释反常现象。

政策分析

政策与数据对齐策略

政策效果的滞后性：关注后几年的数据。

多政策并行分析，经常出现 $A+B+C$ 的叠加情况。
 $A+B+C$ 作为一个输入：政策之间会 $1+1>2$ 。

关注同期未实施类似政策的国家，找到时间趋势。

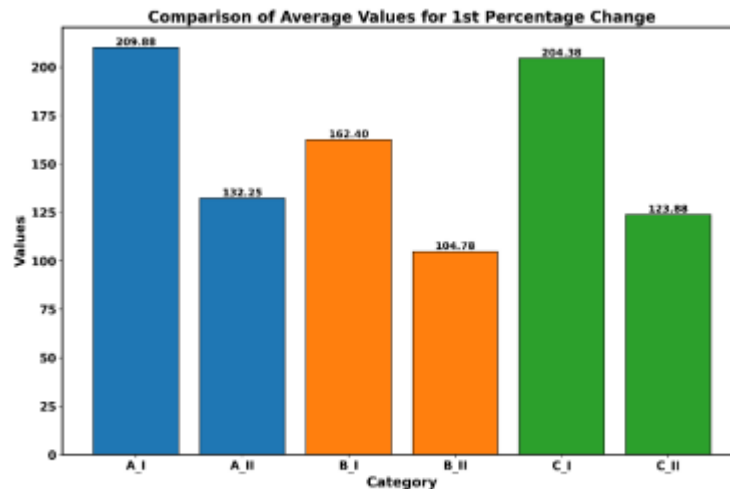
数据处理

Part II

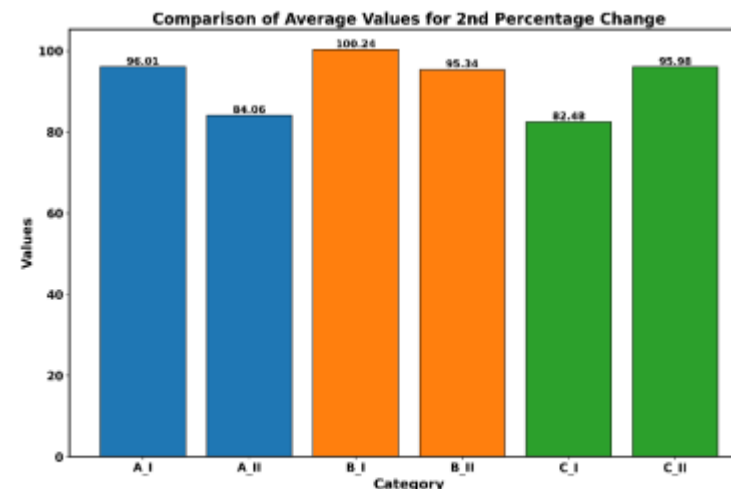
政策收集

政策分析

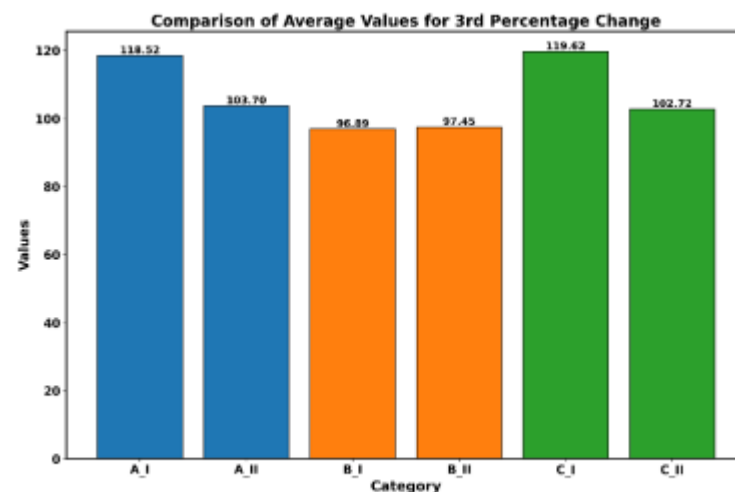
数据处理



(a) The ratio of first year to previous year



(b) The ratio of second year to first year



(c) The ratio of third year to second year

- 生效时间不一致，使用月份分析更精确
- 使用比值进行分析，并非差值
- 数据集缺失：季节权重来分配差额，保证月度的波动

数据收集

相关性分析验证

异常值处理

时序预测验证

数据集来源：
<https://databrowser.uis.unesco.org/>

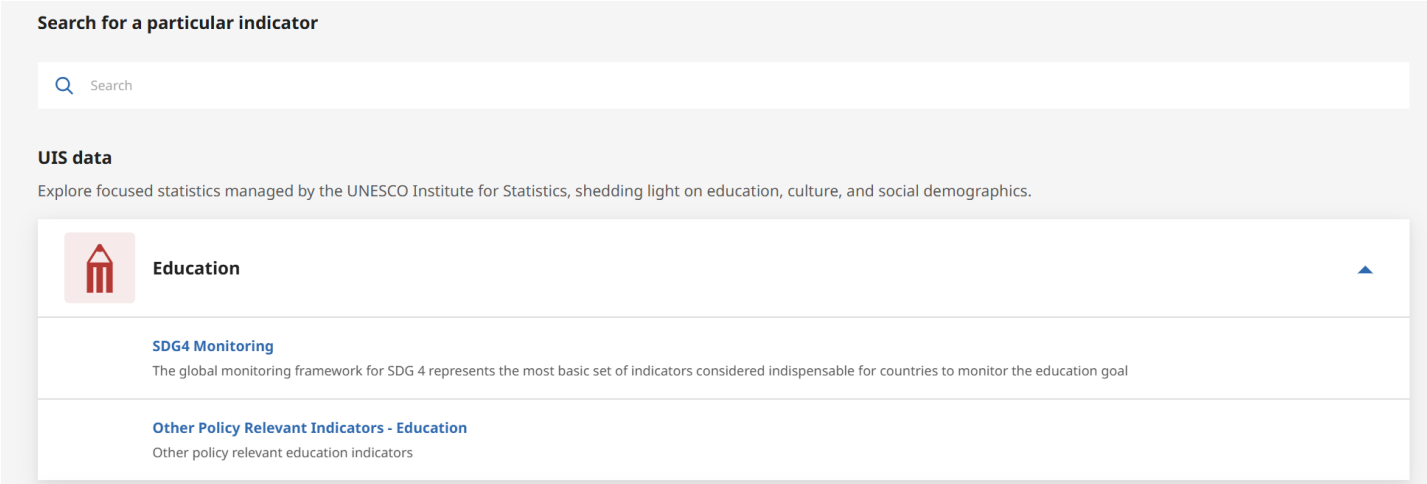


TABLE III. Example dataset for India from 2019-2023

Year	Crime	Enrollment	GDP	Network	Server	Literacy	Education
2019	5	29.0234	2041.4	29.5235	382.86	69	3.8973
2020	21	29.8196	1907.0	43.4054	472.19	69	4.0401
2021	8	31.5742	2239.6	43.4054	528.19	69	4.6351
2022	2	32.6703	2352.6	43.4054	729.69	76	4.1207
2023	1	33.1165	2480.8	43.4054	966.08	77	4.1207

数据搜集

相关性分析验证

异常值处理

时序预测验证

Pearson Correlation Coefficient

$$r = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}}$$

TABLE IV. Result using Pearson Correlation Coefficient in 2023

	Enrollment	GDP	Literacy	Network	Server	Education
Correlation	0.321	0.311	0.249	0.267	-0.099	-0.289

Spearman Rank Correlation Coefficient $\rho = 1 - \frac{6 \sum_{i=1}^n d_i^2}{n(n^2-1)}$

TABLE V. Result using Spearman Rank Correlation Coefficient in 2023

	Enrollment	GDP	Literacy	Network	Server	Education
Correlation	0.348	0.336	0.177	0.306	-0.196	-0.391

Kendall Rank Correlation Coefficient $\tau = \frac{C-D}{\frac{1}{2}n(n-1)}$

TABLE VI. Result using Kendall Rank Correlation Coefficient in 2023

	Enrollment	GDP	Literacy	Network	Server	Education
Correlation	0.299	0.277	0.162	0.254	-0.166	-0.340

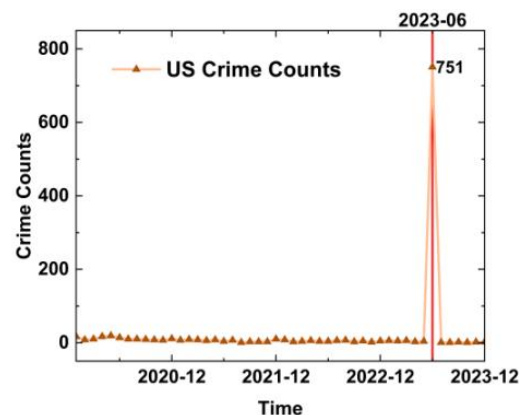
Part III

数据搜集

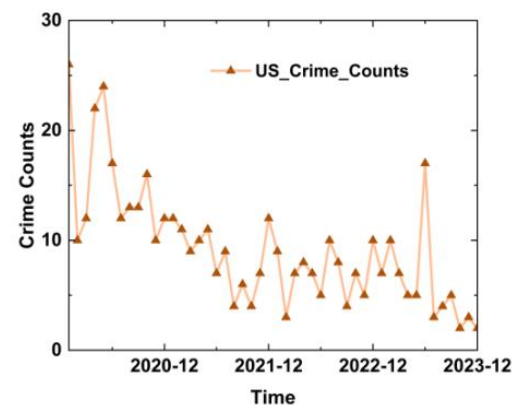
相关性分析验证

异常值处理

时序预测验证



(a) Original data



(b) After correction

```
"actor": {  
  "description": "What entity did the threat action? <a href='http://veriscommunity.net/actors.html' target='_blank'>More Info</a>",
```

文件A

```
"actor": {  
  "external": {  
    "country": [  
      "RU"  
    ],  
    "motive": [  
      "Financial"  
    ],  
    "name": [  
      "Cl0p ransomware group"  
    ],  
    "region": [  
      "150151"  
    ],  
    "variety": [  
      "Organized crime"  
    ],  
  },  
},
```

文件B

```
"actor": {  
  "external": {  
    "country": [  
      "RU"  
    ],  
    "motive": [  
      "Financial"  
    ],  
    "name": [  
      "Cl0p ransomware group"  
    ],  
    "region": [  
      "150151"  
    ],  
    "variety": [  
      "Organized crime"  
    ],  
  },  
},
```

文件C

```
"actor": {  
  "external": {  
    "country": [  
      "RU"  
    ],  
    "motive": [  
      "Financial"  
    ],  
    "name": [  
      "Cl0p ransomware group"  
    ],  
    "region": [  
      "150151"  
    ],  
    "variety": [  
      "Organized crime"  
    ],  
  },  
},
```

完全相同

- Actor
- Asset
- Attribute

不同

- Industry
- Region

其他国家的文件基本不统计此项!

Part III

数据搜集

相关性分析验证

异常值处理

时序预测验证

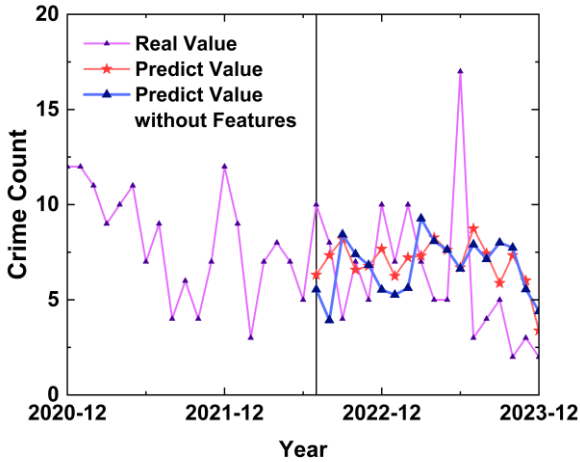
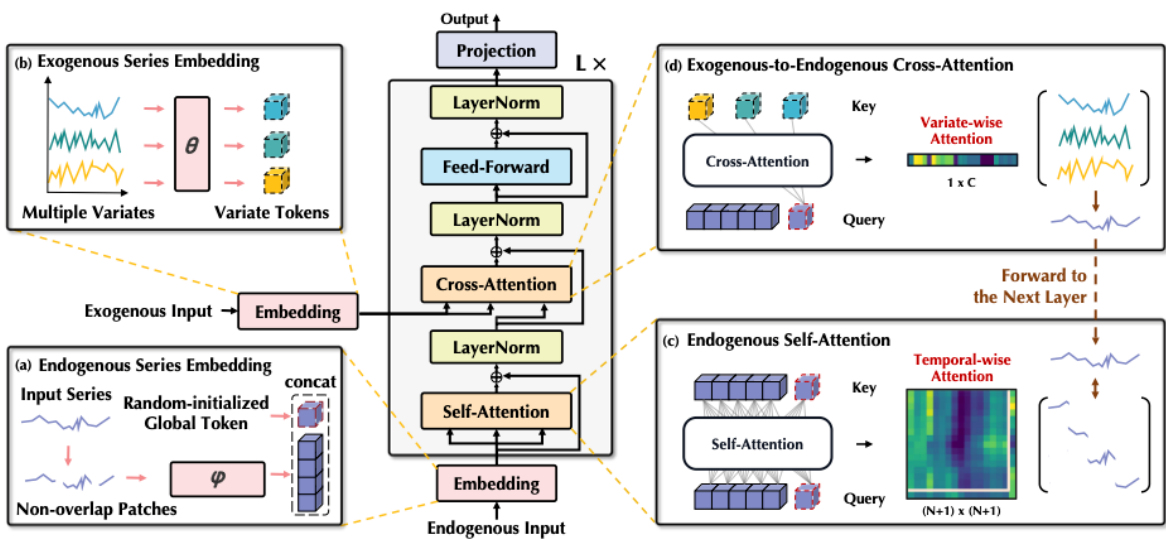


TABLE VII. MAE and MSE for two sets of training

	With Exogenous features	Without Exogenous features
MAE	14.313	17.668
MSE	2.938	3.659

03

备赛建议

结合去年参赛中准备不足的经验，对后续团队的备赛建议



有关MEMO与文件模板

2025 MCM: Memo*3 (B, D, F)

Your PDF solution of no more than 25 total pages should include:

- One-page Summary Sheet.
- Table of Contents.
- Your complete solution.
- One-page memo to the tourist council.
- References list.
- [AI Use Report](#) (If used does not count toward the 25-page limit.)

2024 MCM: Memo*1 (D)

Your PDF solution of no more than 25 pages total should include:

- One-page summary sheet that clearly describes your approach to the problem and your most important conclusions from your analysis in the context of the problem.
- Table of Contents.
- Your complete solution.
- One-page memo.
- Reference List.
- [AI Use Report](#) (if used).

2023 MCM: Non-Technical Report*1 (B)

Your PDF solution of no more than 25 total pages should include:

- One-page Summary Sheet.
- Table of Contents.
- Your complete solution.
- Two-page non-technical report.
- Reference List.

2022 MCM: Letter/Article*4 (B, C, D, E)

Your PDF solution of no more than 25 total pages should include:

- One-page Summary Sheet.
- Table of Contents.
- Your complete solution.
- One- to two-page newspaper article.
- Reference List.

2021 MCM: Document*2 (B, D)

Your PDF solution of no more than 25 total pages should include:

- One-page Summary Sheet.
- Table of Contents.
- Your complete solution.
- One-page document to ICM society.
- References list.

2020 MCM: Article/Letter*3 (A, B, C)

Your submission should consist of:

- One-page Summary Sheet
- Table of Contents
- One- to Two-page Article
- Your solution of no more than 20 pages, for a maximum of 24 pages with your summary, table of contents, and article.

有关MEMO与文件模板

我们的

Team #2517199Page: 25 of 25

Empirical Insights on Cybersecurity: Policy Effectiveness and Predictive Models for Global Cybercrime Trends

While the rapid development of the Internet has driven economic, technological and civilizational progress, it has also facilitated criminal activities such as hacking and DDoS attacks, posing a serious threat to individuals, businesses and governments. Since entering the 21st century, with the rapid expansion of the global Internet user base, governments around the world have introduced various policies to curb the increasingly rampant cybercrime. However, there is still a lack of sufficient empirical studies to measure the real effects of these measures.

Establishing a Rigorous Statistical System for the VCDB Dataset

Using VCDB data compiled by the VERIS community, we systematically analyzed global cybercrime incidents from 1971 to 2023. By combining this data with other relevant indicators, we developed a visualization-based approach to assess the level of risk in different countries. This approach not only provides a clear picture of where and how cybercrime occurs but also supports further research with rich data. However, we identified certain anomalies in the VCDB dataset, including duplicate records. If left unaddressed, these anomalies could skew statistical results, which in turn could undermine research results and policy decisions.

Systematic Evaluation and Ranking of Policy Effectiveness

When comparing countries' cybercrime rates to their GCI scores, we found that countries with high GCI scores do not necessarily have low cybercrime rates. For example, while the U.S. ranked first in the 2020 GCI, the U.S. also had the highest number of recorded cybercrime incidents in the same year, according to VCDB data. This suggests that relying solely on broad indices such as the GCI may not accurately reflect a country's actual cybersecurity posture. To get a more nuanced view of policy effectiveness, we examined several countries with significant changes in cybercrime trends. Based on publicly available government information from these countries, we identified three broad policy directions: institutional development, cooperation, and public education and talent improvement.

A Feature-Fusion-Based Method for Predicting Cybercrime Rates Using National Characteristics

Our findings suggest that pursuing all three types of policies simultaneously is the most effective way to curb cybercrime, resources permitting. If governments can only choose one of these, then partnering with businesses or other countries tends to have the most significant impact. Finally, to predict future trends in cybercrime, we constructed a predictive model that integrates demographic variables that have been shown to correlate with cybercrime rates. Policymakers can use these analyses and predictions to determine whether new security measures should be put in place in order to be more proactive in dealing with the evolving cyberthreats of the digital age



2025 E Outstanding

Topic: Recommendations for Organic Farming Practices
Form: Team # 2508861 of 2025 ICM
Date: January 28, 2025
To our farmer friends
Dear Farmers,

Sustainable organic agriculture can not only effectively control pests, improve soil health, and enhance biodiversity, but also create economic benefits in the ever-expanding organic market. Therefore, we have specially written this letter to provide you with recommendations for exploring organic farming practices.



The transition to organic farming requires initial investments, such as organic fertilizers and biological pest control. These measures effectively control pests, reduce the use of chemicals, and enhance soil fertility, leading to long-term returns. Its sustainability is reflected in improved ecological resilience, enhanced soil productivity, and reduced environmental impact. Therefore, a positive cycle between economics and sustainability requires our collective effort.



有关论文模板:

一定要用Latex?

如果对Word足够熟悉的话，没必要
硬用Latex

注意，是**足够熟悉**！！至少熟练
使用以下操作

- 图片跟随与固定
- 页眉编辑
- 题注引用

有关时序预测与求解器

模型选择基准:

- 一定要前沿 (2024, 2025)
- 推荐选择具有Zero Shot/Few Shot功能的预测模型

例: **UNITS: A Unified Multi-Task Time Series Model**
(不推荐)

项目地址: <https://github.com/mims-harvard/UniTS>

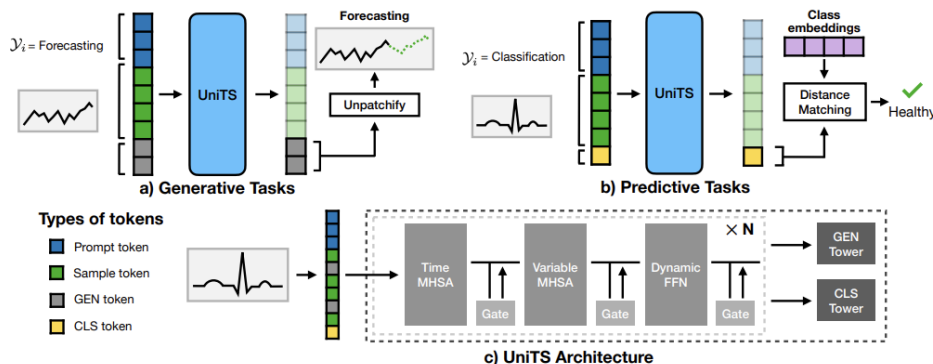


Figure 2: **a)** UNITS for forecasting; input is tokenized, and GEN tokens are un-patchified to infer the forecast horizon. **b)** UNITS for classification; a CLS token is used to represent class information and then compared to class tokens to get prediction class. **c)** Architecture of UNITS model.

模型来源建议:

- NeurIPS
- ICLR
- 或自己实际使用过的项目

求解器选择:

优先选择自己最熟悉的!

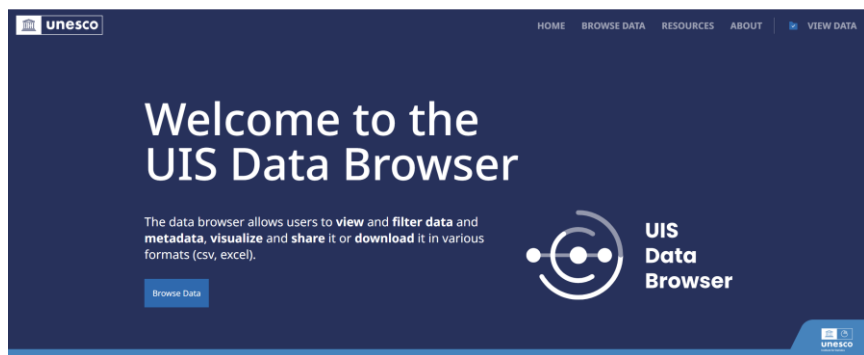
- Python?
- Matlab?
- R语言?
- Lingo?

都不熟悉?

推荐Python---GPT, Claude最擅长

有关数据集网站准备与绘图软件准备

联合国教科文统计网站：**最重要!!!**
<https://databrowser.uis.unesco.org/>



数据集整合网站：
<https://www.selectdataset.com/>



数据集整合网站：
<https://databrowser.uis.unesco.org/>

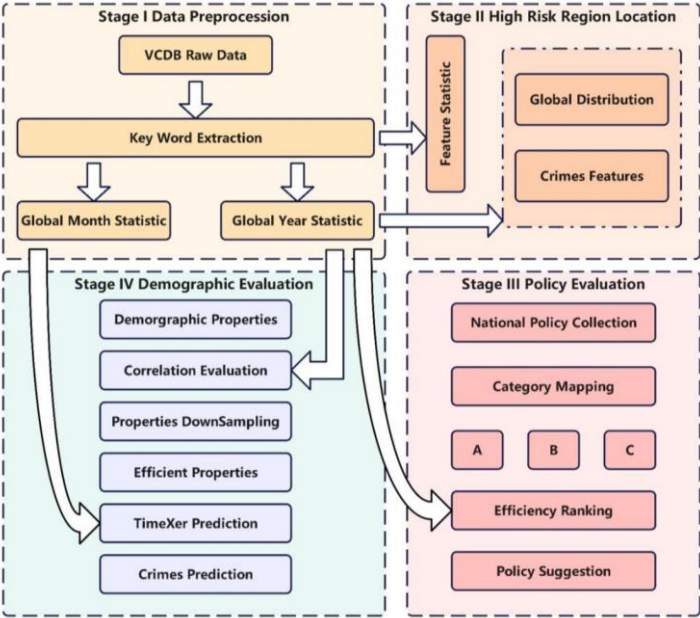


绘图/制表软件：

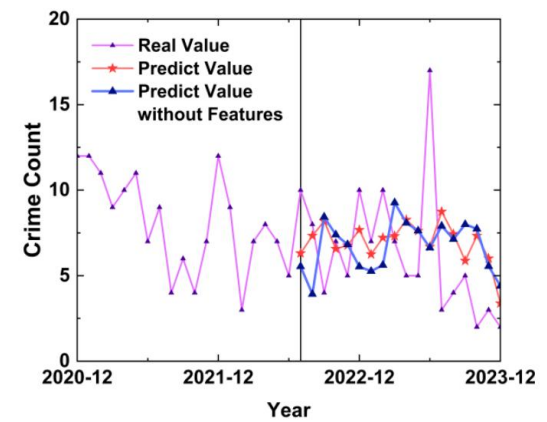
- Origin
- Visio(亿图图示)
- Excel (指表)

需要会画哪种图?

流程图 (Visio)



点线图 (Origin)

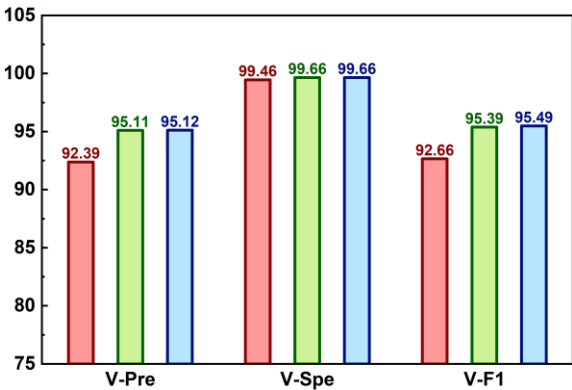


伪代码 (非必要)

```
Algorithm 1 : Distance-Driven Algorithm
Input  : ECG Signal Sequence S (Ordered by Hearbeat)
Output : Predicted Label  $\hat{y}_i$  For Each Hearbeat  $i$ 

For each Hearbeat  $i$  in S
     $(d_{Ni}, d_{Si}, Prompt\ Feature_i) \leftarrow$  First-Stage Process Block ( $S_i$ )
    if  $d_{Ni} < d_{Si} - 2$  then
         $\hat{y}_i \leftarrow N$ 
    else
         $\hat{y}_i \leftarrow \bar{N}$ 
    If  $\bar{N}$  exist in Prediction
        For each Hearbeat  $i$  in S
             $(d_{Ni}, d_{Si}, d_{Vi}, d_{Fi}) \leftarrow$  Second-Stage Process Block( $S_i, Prompt\ Feature_i$ )
             $\hat{y}_i \leftarrow$  CLS-Distance Classification ( $d_{Ni}, d_{Si}, d_{Vi}, d_{Fi}$ )
            Update  $\{\hat{y}_i\}$  with Refined Labels
Return Predicted Label  $\hat{y}_i$  For Each Hearbeat  $i$ 
```

柱状图 (Origin+Visio)



04

赛中建议

结合去年参赛中准备不足的经验，对后续团队的备赛建议



➤ 建模手 一定要是这种分工吗？

➤ 编程手 未必

➤ 论文手

传统分工的好处：

➤ 各司其职

➤ 各取所长

传统分工的坏处（个人想法）：

➤ 时间上存在浪费，无法流水线工作

➤ 算法沟通上容易存在问题

我们的分工方式：

三人共同参与选题的讨论并制定初步的研究规划

没有明确的建模，编程，论文的分工界限，大家互相承担一部分对方的内容

两人主要负责建模与求解，一人主要负责整体论文的排版与整理

流水线工作，以此次2025 F题为例，Part I与Part II的工作在前期可以并行处理

为什么选择这种分工：

AI时代，只要会写提示词，能看懂代码，就足够了，不需要会写代码

AI可以帮助英文翻译，只要对翻译后的文本进行修改就可以了

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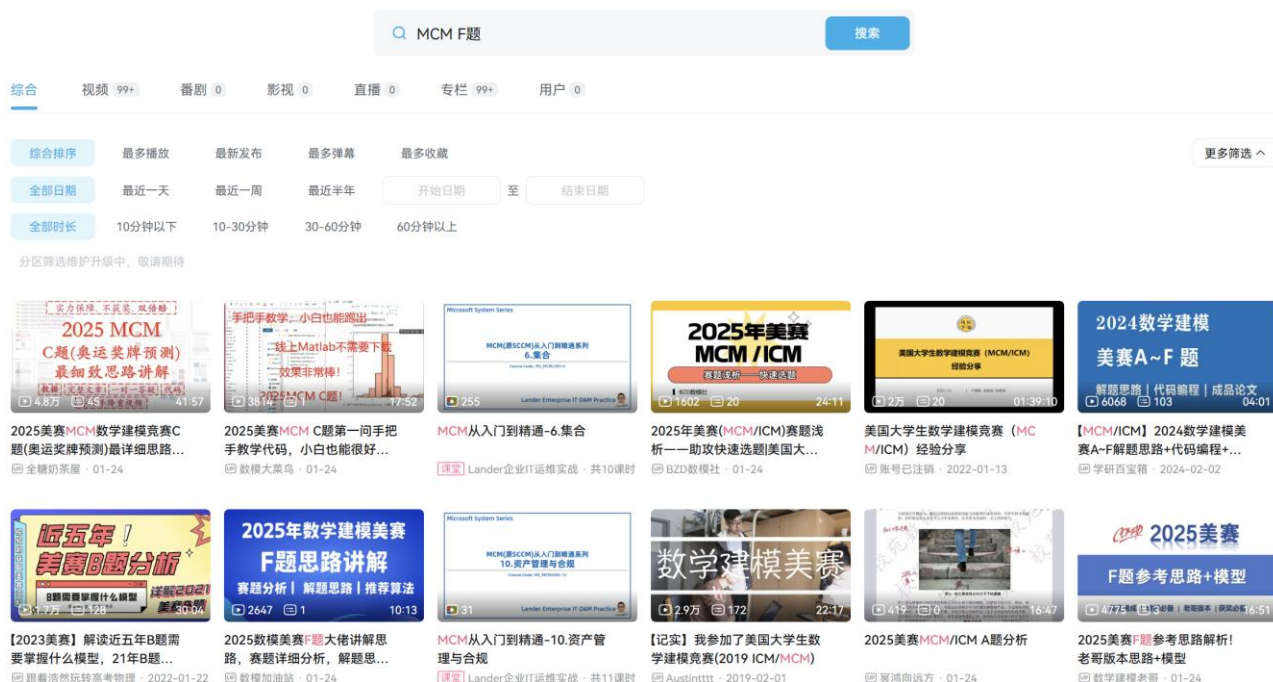
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最重要!!!
选择适合团队的分工方式

有关AI与网上资源

网上资源：BiliBili，CSDN，知乎



个人建议:

首先团队内部讨论，完成选题+解题思路构建，不要看任何网上资源也不要使用任何AI工具（除翻译）
---避免被外部混淆思考

可以在完成解题大纲规划后，让AI根据赛题给出一份解题大纲，同时可以参考下网上资源给出的解题思路，并对团队内提前指定的解题思路进行对比与调整
---注意，是**参考**，而不是**照抄**

对于每一题的求解过程，不要吝啬使用AI工具，这将大大加快求解效率，尤其是在资源搜索，编程求解上
---但是不建议在建模上依赖AI，可以仅作参考

AI资源:

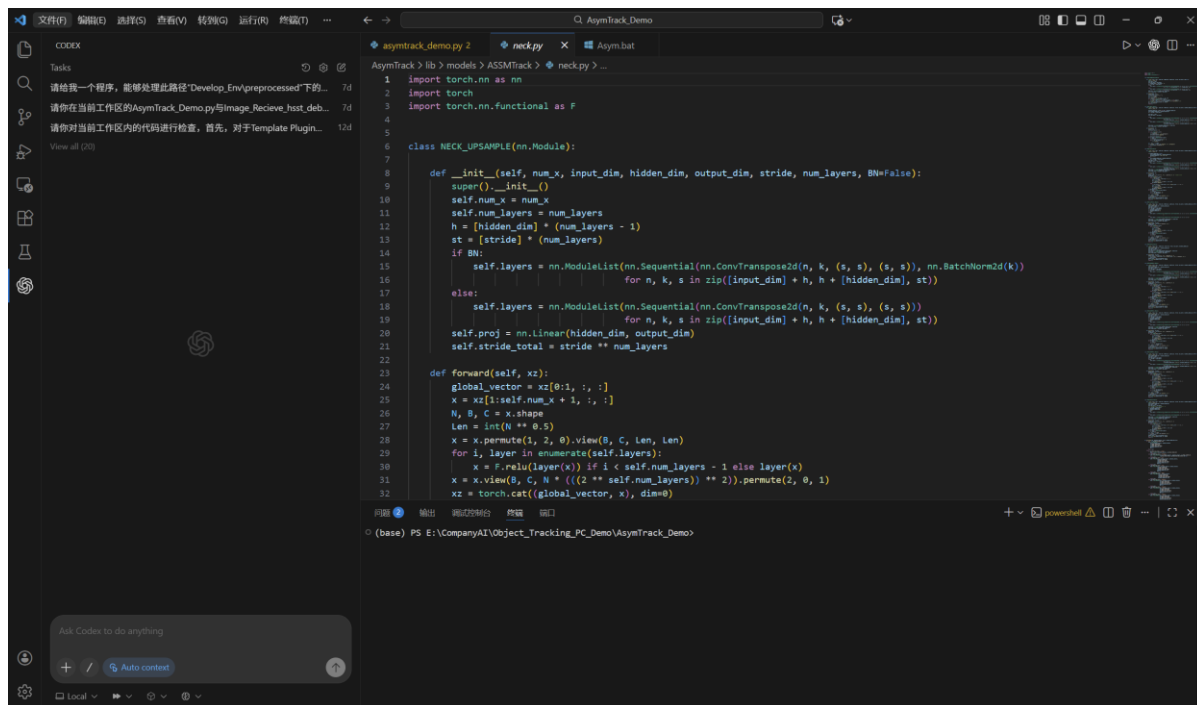
GPT(Codex), Claude---编程

GPT, Gemini---文本, 搜索, 思考

建议使用Vscode，并安装AI辅助插件---省略对文件存储逻辑等步骤描述的提示词，AI可以自己看工作区

有关AI与网上资源

网上资源：BiliBili，CSDN，知乎



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有关AI与网上资源

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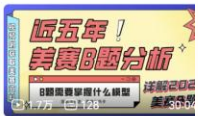
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MM-Agent: LLM as Agents for Real-world Mathematical Modeling Problem

Fan Liu, Zhe-Rui Yang, Cancheng Liu, Tianrui SONG, Xiaofeng Gao, Hao Liu

Published: 19 Sept 2025, Last Modified: 29 Oct 2025 NeurIPS 2025 poster Everyone Revisions BibTeX CC BY-NC-ND 4.0

Keywords: Mathematical Modeling Agent, LLM Agent, LLM

TL;DR: We introduce MM-Agent and MM-Bench to enable and evaluate LLM-powered end-to-end mathematical modeling, achieving superior performance on real-world problems and Finalist Award in MCM/ICM 2025.

Abstract:

Mathematical modeling is a cornerstone of scientific discovery and engineering practice, enabling the translation of real-world problems into formal systems across domains such as physics, biology, and economics. Unlike mathematical reasoning, which assumes a predefined formulation, modeling requires open-ended problem analysis, abstraction, and principled formalization. While Large Language Models (LLMs) have shown strong reasoning capabilities, they fall short in rigorous model construction, limiting their utility in real-world problem-solving. To this end, we formalize the task of LLM-powered real-world mathematical modeling, where agents must analyze problems, construct domain-appropriate formulations, and generate complete end-to-end solutions. We introduce MM-Bench, a curated benchmark of 111 problems from the Mathematical Contest in Modeling (MCM/ICM), spanning the years 2000 to 2025 and across ten diverse domains such as physics, biology, and economics. To tackle this task, we propose MM-Agent, an expert-inspired framework that decomposes mathematical modeling into four stages: open-ended problem analysis, structured model formulation, computational problem solving, and report generation. Experiments on MM-Bench show that MM-Agent significantly outperforms baseline agents, achieving an 11.88% improvement over human expert solutions while requiring only 15 minutes and \$0.88 per task using GPT-4o. Furthermore, under official MCM/ICM protocols, MM-Agent assisted two undergraduate teams in winning the Finalist Award (textbf{top 2.0% among 27,456 teams}) in MCM/ICM 2025, demonstrating its practical effectiveness as a modeling copilot.

Primary Area: Deep learning (e.g., architectures, generative models, optimization for deep networks, foundation models, LLMs)

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要使用

AI根据
以参考
对团队
与调整

豈使用
，尤其

可以仅

AI资源：
GPT(Co
GPT，G
建议使用
辑等步骤



THANKS

汇报人：王翊钊，赵浩安

联系方式：19808147932